PG No. 40 $C_{4h}(c)$ $4/m$ [tetragonal] (internal axial quadrupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(2,2)}[g](A_u)$

** symmetry

1

** expression

$$-\frac{\sqrt{5}G_u\left(x^2+y^2-2z^2\right)}{10}+\frac{\sqrt{15}G_v\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{15}G_{xy}xy}{5}+\frac{\sqrt{15}G_{xz}xz}{5}+\frac{\sqrt{15}G_{yz}yz}{5}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(2,0)}[g](A_g)$

** symmetry

z

** expression

$$\frac{\sqrt{10}G_u z}{5}+\frac{\sqrt{30}G_{xz}x}{10}+\frac{\sqrt{30}G_{yz}y}{10}$$

$\vec{\mathbb{G}}_1^{(2,2)}[g](A_g)$

** symmetry

z

** expression

$$-\frac{3\sqrt{35}G_u z\left(3x^2+3y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_v z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_{xy}xyz}{7}-\frac{\sqrt{105}G_{xz}x\left(x^2+y^2-4z^2\right)}{35}-\frac{\sqrt{105}G_{yz}y\left(x^2+y^2-4z^2\right)}{35}$$

$\vec{\mathbb{G}}_1^{(2,0)}[g](E_g^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{5}G_u\left(x+iy\right)}{10}+\frac{\sqrt{15}G_v\left(x-iy\right)}{10}+\frac{\sqrt{15}iG_{xy}\left(x-iy\right)}{10}+\frac{\sqrt{15}G_{xz}z}{10}+\frac{\sqrt{15}iG_{yz}z}{10}$$

$\vec{\mathbb{G}}_1^{(2,2)}[g](E_g^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{70}G_u\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{140}+\frac{\sqrt{210}G_v\left(3x^3+7ix^2y-7xy^2-2xz^2-3iy^3+2iyz^2\right)}{140}$$
$$-\frac{\sqrt{210}iG_{xy}\left(x^3+4ix^2y-4xy^2+xz^2-iy^3-iyz^2\right)}{70}+\frac{\sqrt{210}G_{xz}z\left(4x^2+5ixy-y^2-z^2\right)}{70}-\frac{\sqrt{210}iG_{yz}z\left(x^2+5ixy-4y^2+z^2\right)}{70}$$

$\vec{\mathbb{G}}_1^{(2,0)}[g](E_g^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{5}G_u\left(x-iy\right)}{10}+\frac{\sqrt{15}G_v\left(x+iy\right)}{10}-\frac{\sqrt{15}iG_{xy}\left(x+iy\right)}{10}+\frac{\sqrt{15}G_{xz}z}{10}-\frac{\sqrt{15}iG_{yz}z}{10}$$

$\vec{\mathbb{G}}_1^{(2,2)}[g](E_g^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{70}G_u\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{140}+\frac{\sqrt{210}G_v\left(3x^3-7ix^2y-7xy^2-2xz^2+3iy^3-2iyz^2\right)}{140}$$
$$+\frac{\sqrt{210}iG_{xy}\left(x^3-4ix^2y-4xy^2+xz^2+iy^3+iyz^2\right)}{70}+\frac{\sqrt{210}G_{xz}z\left(4x^2-5ixy-y^2-z^2\right)}{70}+\frac{\sqrt{210}iG_{yz}z\left(x^2-5ixy-4y^2+z^2\right)}{70}$$

$\vec{\mathbb{Q}}_1^{(2,1)}[g](A_u)$

** symmetry

z

** expression

$$\frac{\sqrt{30}G_v xy}{5} - \frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{10} + \frac{\sqrt{30}G_{xz}yz}{10} - \frac{\sqrt{30}G_{yz}xz}{10}$$

$\vec{\mathbb{Q}}_1^{(2,1)}[g](E_u^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{3\sqrt{5}iG_u z\left(x+iy\right)}{10} - \frac{\sqrt{15}iG_v z\left(x-iy\right)}{10} + \frac{\sqrt{15}G_{xy}z\left(x-iy\right)}{10} + \frac{\sqrt{15}iG_{xz}\left(x^2+ixy-z^2\right)}{10} + \frac{\sqrt{15}iG_{yz}\left(xy+iy^2-iz^2\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(2,1)}[g](E_u^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{5}iG_u z\left(x-iy\right)}{10} + \frac{\sqrt{15}iG_v z\left(x+iy\right)}{10} + \frac{\sqrt{15}G_{xy}z\left(x+iy\right)}{10} - \frac{\sqrt{15}iG_{xz}\left(x^2-ixy-z^2\right)}{10} - \frac{\sqrt{15}iG_{yz}\left(xy-iy^2+iz^2\right)}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{2}G_{xz}y}{2} - \frac{\sqrt{2}G_{yz}x}{2}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{5\sqrt{42}G_v xyz}{14} - \frac{5\sqrt{42}G_{xy}z\left(x-y\right)\left(x+y\right)}{28} - \frac{\sqrt{42}G_{xz}y\left(x^2+y^2-4z^2\right)}{28} + \frac{\sqrt{42}G_{yz}x\left(x^2+y^2-4z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](B_g,1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{6}G_{xy}z}{3} - \frac{\sqrt{6}G_{xz}y}{6} - \frac{\sqrt{6}G_{yz}x}{6}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](B_g,2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{6}G_v z}{3} + \frac{\sqrt{6}G_{xz}x}{6} - \frac{\sqrt{6}G_{yz}y}{6}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](B_g,1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{5\sqrt{42}G_u xyz}{14}+\frac{\sqrt{14}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{28}-\frac{\sqrt{14}G_{xz}y\left(9x^2-y^2-6z^2\right)}{28}+\frac{\sqrt{14}G_{yz}x\left(x^2-9y^2+6z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](B_g,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{42}G_u z\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{14}G_v z\left(3x^2+3y^2-2z^2\right)}{28}+\frac{\sqrt{14}G_{xz}x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{14}G_{yz}y\left(3x^2-2y^2+3z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](E_g^{(a)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{G_u\left(x+iy\right)}{2}+\frac{\sqrt{3}G_v\left(x-iy\right)}{6}+\frac{\sqrt{3}iG_{xy}\left(x-iy\right)}{6}-\frac{\sqrt{3}G_{xz}z}{6}-\frac{\sqrt{3}iG_{yz}z}{6}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](E_g^{(a)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{21}G_u\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{7}G_v\left(x^3+9ix^2y-9xy^2+6xz^2-iy^3-6iyz^2\right)}{28}$$
$$+\frac{\sqrt{7}iG_{xy}\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{14}+\frac{\sqrt{7}G_{xz}z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{7}iG_{yz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](E_g^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{G_u\left(x-iy\right)}{2}+\frac{\sqrt{3}G_v\left(x+iy\right)}{6}-\frac{\sqrt{3}iG_{xy}\left(x+iy\right)}{6}-\frac{\sqrt{3}G_{xz}z}{6}+\frac{\sqrt{3}iG_{yz}z}{6}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](E_g^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{21}G_u\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{7}G_v\left(x^3-9ix^2y-9xy^2+6xz^2+iy^3+6iyz^2\right)}{28}$$
$$-\frac{\sqrt{7}iG_{xy}\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{14}+\frac{\sqrt{7}G_{xz}z\left(3x^2+3y^2-2z^2\right)}{14}-\frac{\sqrt{7}iG_{yz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$G_u$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{14}G_u\left(x^2+y^2-2z^2\right)}{14}-\frac{\sqrt{42}G_v\left(x-y\right)\left(x+y\right)}{14}-\frac{\sqrt{42}G_{xy}xy}{7}+\frac{\sqrt{42}G_{xz}xz}{14}+\frac{\sqrt{42}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](A_u)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{14}G_u\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{56}-\frac{5\sqrt{42}G_v\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}$$
$$-\frac{5\sqrt{42}G_{xy}xy\left(x^2+y^2-6z^2\right)}{84}-\frac{5\sqrt{42}G_{xz}xz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{42}G_{yz}yz\left(3x^2+3y^2-4z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](B_u,1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$G_v$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](B_u,2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$G_{xy}$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](B_u,1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{42}G_u\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{14}G_v\left(x^2+y^2-2z^2\right)}{14}+\frac{3\sqrt{14}G_{xz}xz}{14}-\frac{3\sqrt{14}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](B_u,2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_uxy}{7}+\frac{\sqrt{14}G_{xy}\left(x^2+y^2-2z^2\right)}{14}+\frac{3\sqrt{14}G_{xz}yz}{14}+\frac{3\sqrt{14}G_{yz}xz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](B_u,1)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{5\sqrt{42}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}+\frac{\sqrt{14}G_v\left(19x^4-102x^2y^2-12x^2z^2+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$+\frac{5\sqrt{14}G_{xy}xy\left(x-y\right)\left(x+y\right)}{12}+\frac{5\sqrt{14}G_{xz}xz\left(5x^2-9y^2-2z^2\right)}{84}+\frac{5\sqrt{14}G_{yz}yz\left(9x^2-5y^2+2z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](B_u,2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{5\sqrt{42}G_uxy\left(x^2+y^2-6z^2\right)}{84}+\frac{5\sqrt{14}G_vxy\left(x-y\right)\left(x+y\right)}{12}-\frac{\sqrt{14}G_{xy}\left(4x^4-27x^2y^2+3x^2z^2+4y^4+3y^2z^2-z^4\right)}{42}$$
$$+\frac{5\sqrt{14}G_{xz}yz\left(6x^2-y^2-z^2\right)}{42}-\frac{5\sqrt{14}G_{yz}xz\left(x^2-6y^2+z^2\right)}{42}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](E_u^{(a)})$

** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{2}iG_{xz}}{2}+\frac{\sqrt{2}G_{yz}}{2}$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](E_u^{(a)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{\sqrt{21}iG_uz\left(x+iy\right)}{14}-\frac{3\sqrt{7}iG_vz\left(x-iy\right)}{14}+\frac{3\sqrt{7}G_{xy}z\left(x-iy\right)}{14}-\frac{\sqrt{7}iG_{xz}\left(x^2+3ixy-2y^2+z^2\right)}{14}-\frac{\sqrt{7}G_{yz}\left(2x^2+3ixy-y^2-z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](E_u^{(a)})$
** symmetry

$$-\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{5\sqrt{21}iG_uz\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{7}iG_vz\left(5x^3+9ix^2y-9xy^2-2xz^2-5iy^3+2iyz^2\right)}{84}$$
$$-\frac{5\sqrt{7}G_{xy}z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{42}+\frac{\sqrt{7}iG_{xz}\left(4x^4+5ix^3y+3x^2y^2-27x^2z^2+5ixy^3-30ixyz^2-y^4+3y^2z^2+4z^4\right)}{42}$$
$$+\frac{\sqrt{7}G_{yz}\left(x^4+5ix^3y-3x^2y^2-3x^2z^2+5ixy^3-30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](E_u^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{2}iG_{xz}}{2}+\frac{\sqrt{2}G_{yz}}{2}$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](E_u^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$\frac{\sqrt{21}iG_uz\left(x-iy\right)}{14}+\frac{3\sqrt{7}iG_vz\left(x+iy\right)}{14}+\frac{3\sqrt{7}G_{xy}z\left(x+iy\right)}{14}+\frac{\sqrt{7}iG_{xz}\left(x^2-3ixy-2y^2+z^2\right)}{14}-\frac{\sqrt{7}G_{yz}\left(2x^2-3ixy-y^2-z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](E_u^{(b)})$
** symmetry

$$\frac{\sqrt{6}ixz}{2}+\frac{\sqrt{6}yz}{2}$$

** expression

$$-\frac{5\sqrt{21}iG_uz\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{7}iG_vz\left(5x^3-9ix^2y-9xy^2-2xz^2+5iy^3-2iyz^2\right)}{84}$$
$$-\frac{5\sqrt{7}G_{xy}z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{42}-\frac{\sqrt{7}iG_{xz}\left(4x^4-5ix^3y+3x^2y^2-27x^2z^2-5ixy^3+30ixyz^2-y^4+3y^2z^2+4z^4\right)}{42}$$
$$+\frac{\sqrt{7}G_{yz}\left(x^4-5ix^3y-3x^2y^2-3x^2z^2-5ixy^3+30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_3^{(2,-2)}[g](A_g)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}G_u z}{5} - \frac{\sqrt{5}G_{xz}x}{5} - \frac{\sqrt{5}G_{yz}y}{5}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](A_g)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{15}G_u z\left(3x^2+3y^2-2z^2\right)}{15} - \frac{\sqrt{5}G_v z\left(x-y\right)\left(x+y\right)}{2} - \sqrt{5}G_{xy}xyz - \frac{\sqrt{5}G_{xz}x\left(x^2+y^2-4z^2\right)}{20} - \frac{\sqrt{5}G_{yz}y\left(x^2+y^2-4z^2\right)}{20}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](A_g)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{330}G_u z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264} - \frac{7\sqrt{110}G_v z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88} - \frac{7\sqrt{110}G_{xy}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+ \frac{\sqrt{110}G_{xz}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44} + \frac{\sqrt{110}G_{yz}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_g,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z}{3} + \frac{\sqrt{3}G_{xz}y}{3} + \frac{\sqrt{3}G_{yz}x}{3}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_g,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_v z}{3} + \frac{\sqrt{3}G_{xz}x}{3} - \frac{\sqrt{3}G_{yz}y}{3}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_g,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{6} + \frac{\sqrt{3}G_{xz}y\left(3x^2-2y^2+3z^2\right)}{6} - \frac{\sqrt{3}G_{yz}x\left(2x^2-3y^2-3z^2\right)}{6}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_g,2)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_v z\left(3x^2+3y^2-2z^2\right)}{6} + \frac{\sqrt{3}G_{xz}x\left(x^2-9y^2+6z^2\right)}{12} + \frac{\sqrt{3}G_{yz}y\left(9x^2-y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_g,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{21\sqrt{22}G_u xyz\left(x^2+y^2-2z^2\right)}{44} + \frac{21\sqrt{66}G_v xyz\left(x-y\right)\left(x+y\right)}{44} - \frac{\sqrt{66}G_{xy}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{66}$$
$$- \frac{\sqrt{66}G_{xz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{66} + \frac{\sqrt{66}G_{yz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{66}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_g, 2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{22}G_u z\,(x-y)\,(x+y)\left(x^2+y^2-2z^2\right)}{88} + \frac{\sqrt{66}G_v z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{264} + \frac{21\sqrt{66}G_{xy}xyz\,(x-y)\,(x+y)}{44}$$
$$-\frac{\sqrt{66}G_{xz}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{132} - \frac{\sqrt{66}G_{yz}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{132}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](E_g^{(a)}, 1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2} - \frac{3\sqrt{2}ix^2y}{4} - \frac{3\sqrt{2}xy^2}{4} - \frac{3\sqrt{2}xz^2}{4} + \frac{\sqrt{2}iy^3}{2} - \frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{30}G_u\,(x+iy)}{20} + \frac{3\sqrt{10}G_v\,(x-iy)}{20} - \frac{\sqrt{10}iG_{xy}\,(x-iy)}{10} - \frac{\sqrt{10}G_{xz}z}{10} - \frac{\sqrt{10}iG_{yz}z}{10}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](E_g^{(a)}, 2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4} + \frac{\sqrt{30}xy^2}{4} - \frac{\sqrt{30}xz^2}{4} - \frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{2}G_u\,(x+iy)}{4} - \frac{\sqrt{6}G_v\,(x-iy)}{12} + \frac{\sqrt{6}iG_{xy}\,(x-iy)}{6} - \frac{\sqrt{6}G_{xz}z}{6} - \frac{\sqrt{6}iG_{yz}z}{6}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](E_g^{(a)}, 1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2} - \frac{3\sqrt{2}ix^2y}{4} - \frac{3\sqrt{2}xy^2}{4} - \frac{3\sqrt{2}xz^2}{4} + \frac{\sqrt{2}iy^3}{2} - \frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{30}G_u\left(4x^3-21ix^2y-21xy^2+9xz^2+4iy^3+9iyz^2\right)}{120} + \frac{\sqrt{10}G_v\left(4x^3+ix^2y-xy^2-11xz^2-4iy^3+11iyz^2\right)}{40}$$
$$-\frac{\sqrt{10}iG_{xy}\left(x^3+4ix^2y-4xy^2+xz^2-iy^3-iyz^2\right)}{40} + \frac{\sqrt{10}G_{xz}z\left(4x^2-20ixy-y^2-z^2\right)}{40} - \frac{\sqrt{10}iG_{yz}z\left(x^2-20ixy-4y^2+z^2\right)}{40}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](E_g^{(a)}, 2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4} + \frac{\sqrt{30}xy^2}{4} - \frac{\sqrt{30}xz^2}{4} - \frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_u\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{8} + \frac{\sqrt{6}G_v\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{24}$$
$$+\frac{\sqrt{6}iG_{xy}\left(x^3-6ix^2y+6xy^2-9xz^2-iy^3+9iyz^2\right)}{24} - \frac{\sqrt{6}G_{xz}z\left(6x^2-9y^2+z^2\right)}{24} + \frac{\sqrt{6}iG_{yz}z\left(9x^2-6y^2-z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](E_g^{(a)}, 1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2} - \frac{3\sqrt{2}ix^2y}{4} - \frac{3\sqrt{2}xy^2}{4} - \frac{3\sqrt{2}xz^2}{4} + \frac{\sqrt{2}iy^3}{2} - \frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{165}G_u\left(4x^5-3ix^4y+x^3y^2-41x^3z^2+ix^2y^3+15ix^2yz^2-3xy^4+15xy^2z^2+18xz^4+4iy^5-41iy^3z^2+18iyz^4\right)}{264}$$
$$+\frac{\sqrt{55}G_v\left(4x^5-11ix^4y-27x^3y^2-13x^3z^2+27ix^2y^3-15ix^2yz^2+11xy^4+15xy^2z^2+4xz^4-4iy^5+13iy^3z^2-4iyz^4\right)}{88}$$
$$+\frac{\sqrt{55}iG_{xy}\left(x^5-8ix^4y-12x^3y^2+2x^3z^2+12ix^2y^3+12ix^2yz^2+8xy^4-12xy^2z^2+xz^4-iy^5-2iy^3z^2-iyz^4\right)}{44}$$
$$+\frac{\sqrt{55}G_{xz}z\left(8x^4-7ix^3y-12x^2y^2-12x^2z^2+14ixy^3-7ixyz^2+y^4+2y^2z^2+z^4\right)}{44}$$
$$+\frac{\sqrt{55}iG_{yz}z\left(x^4-14ix^3y-12x^2y^2+2x^2z^2+7ixy^3+7ixyz^2+8y^4-12y^2z^2+z^4\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](E_g^{(a)},2)$

** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{11}G_u\left(2x^5+9ix^4y+11x^3y^2-31x^3z^2+11ix^2y^3-87ix^2yz^2+9xy^4-87xy^2z^2+30xz^4+2iy^5-31iy^3z^2+30iyz^4\right)}{88}$$
$$-\frac{\sqrt{33}G_v\left(2x^5-51ix^4y-73x^3y^2+53x^3z^2+73ix^2y^3+87ix^2yz^2+51xy^4-87xy^2z^2-12xz^4-2iy^5-53iy^3z^2+12iyz^4\right)}{264}$$
$$-\frac{\sqrt{33}iG_{xy}\left(5x^5-12ix^4y-46x^3y^2-4x^3z^2+46ix^2y^3-66ix^2yz^2+12xy^4+66xy^2z^2-9xz^4-5iy^5+4iy^3z^2+9iyz^4\right)}{132}$$
$$+\frac{\sqrt{33}G_{xz}z\left(12x^4+63ix^3y+66x^2y^2-46x^2z^2-63ixyz^2-9y^4-4y^2z^2+5z^4\right)}{132}$$
$$-\frac{\sqrt{33}iG_{yz}z\left(9x^4-66x^2y^2+4x^2z^2+63ixy^3-63ixyz^2-12y^4+46y^2z^2-5z^4\right)}{132}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](E_g^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{30}G_u\left(x-iy\right)}{20}+\frac{3\sqrt{10}G_v\left(x+iy\right)}{20}+\frac{\sqrt{10}iG_{xy}\left(x+iy\right)}{10}-\frac{\sqrt{10}G_{xz}z}{10}+\frac{\sqrt{10}iG_{yz}z}{10}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](E_g^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{2}G_u\left(x-iy\right)}{4}-\frac{\sqrt{6}G_v\left(x+iy\right)}{12}-\frac{\sqrt{6}iG_{xy}\left(x+iy\right)}{6}-\frac{\sqrt{6}G_{xz}z}{6}+\frac{\sqrt{6}iG_{yz}z}{6}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](E_g^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{30}G_u\left(4x^3+21ix^2y-21xy^2+9xz^2-4iy^3-9iyz^2\right)}{120}+\frac{\sqrt{10}G_v\left(4x^3-ix^2y-xy^2-11xz^2+4iy^3-11iyz^2\right)}{40}$$
$$+\frac{\sqrt{10}iG_{xy}\left(x^3-4ix^2y-4xy^2+xz^2+iy^3+iyz^2\right)}{40}+\frac{\sqrt{10}G_{xz}z\left(4x^2+20ixy-y^2-z^2\right)}{40}+\frac{\sqrt{10}iG_{yz}z\left(x^2+20ixy-4y^2+z^2\right)}{40}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](E_g^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{2}G_u\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{8}+\frac{\sqrt{6}G_v\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{24}$$
$$-\frac{\sqrt{6}iG_{xy}\left(x^3+6ix^2y+6xy^2-9xz^2+iy^3-9iyz^2\right)}{24}-\frac{\sqrt{6}G_{xz}z\left(6x^2-9y^2+z^2\right)}{24}-\frac{\sqrt{6}iG_{yz}z\left(9x^2-6y^2-z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](E_g^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{165}G_u\left(4x^5+3ix^4y+x^3y^2-41x^3z^2-ix^2y^3-15ix^2yz^2-3xy^4+15xy^2z^2+18xz^4-4iy^5+41iy^3z^2-18iyz^4\right)}{264}$$
$$+\frac{\sqrt{55}G_v\left(4x^5+11ix^4y-27x^3y^2-13x^3z^2-27ix^2y^3+15ix^2yz^2+11xy^4+15xy^2z^2+4xz^4+4iy^5-13iy^3z^2+4iyz^4\right)}{88}$$
$$-\frac{\sqrt{55}iG_{xy}\left(x^5+8ix^4y-12x^3y^2+2x^3z^2-12ix^2y^3-12ix^2yz^2+8xy^4-12xy^2z^2+xz^4+iy^5+2iy^3z^2+iyz^4\right)}{44}$$
$$+\frac{\sqrt{55}G_{xz}z\left(8x^4+7ix^3y-12x^2y^2-12x^2z^2-14ixy^3+7ixyz^2+y^4+2y^2z^2+z^4\right)}{44}$$
$$-\frac{\sqrt{55}iG_{yz}z\left(x^4+14ix^3y-12x^2y^2+2x^2z^2-7ixy^3-7ixyz^2+8y^4-12y^2z^2+z^4\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](E_g^{(b)}, 2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{11}G_u\left(2x^5-9ix^4y+11x^3y^2-31x^3z^2-11ix^2y^3+87ix^2yz^2+9xy^4-87xy^2z^2+30xz^4-2iy^5+31iy^3z^2-30iyz^4\right)}{88}$$
$$-\frac{\sqrt{33}G_v\left(2x^5+51ix^4y-73x^3y^2+53x^3z^2-73ix^2y^3-87ix^2yz^2+51xy^4-87xy^2z^2-12xz^4+2iy^5+53iy^3z^2-12iyz^4\right)}{264}$$
$$+\frac{\sqrt{33}iG_{xy}\left(5x^5+12ix^4y-46x^3y^2-4x^3z^2-46ix^2y^3+66ix^2yz^2+12xy^4+66xy^2z^2-9xz^4+5iy^5-4iy^3z^2-9iyz^4\right)}{132}$$
$$+\frac{\sqrt{33}G_{xz}z\left(12x^4-63ix^3y+66x^2y^2-46x^2z^2+63ixyz^2-9y^4-4y^2z^2+5z^4\right)}{132}$$
$$+\frac{\sqrt{33}iG_{yz}z\left(9x^4-66x^2y^2+4x^2z^2-63ixy^3+63ixyz^2-12y^4+46y^2z^2-5z^4\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](A_u)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{30}G_vxy}{10}+\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{20}+\frac{\sqrt{30}G_{xz}yz}{5}-\frac{\sqrt{30}G_{yz}xz}{5}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](A_u)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_vxy\left(x^2+y^2-6z^2\right)}{6}+\frac{\sqrt{15}G_{xy}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}-\frac{\sqrt{15}G_{xz}yz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{\sqrt{15}G_{yz}xz\left(3x^2+3y^2-4z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](B_u, 1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{6}G_u\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{2}G_v\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](B_u, 2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}G_uxy}{2}-\frac{\sqrt{2}G_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_u, 1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{6}-\frac{G_v\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{6}$$
$$-\frac{7G_{xy}xy\left(x-y\right)\left(x+y\right)}{6}+\frac{7G_{xz}xz\left(x-z\right)\left(x+z\right)}{6}-\frac{7G_{yz}yz\left(y-z\right)\left(y+z\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_u, 2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_u xy\left(x^2+y^2-6z^2\right)}{3}+\frac{7G_v xy\left(x-y\right)\left(x+y\right)}{6}-\frac{G_{xy}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{12}$$
$$-\frac{7G_{xz}yz\left(3x^2+y^2-2z^2\right)}{12}-\frac{7G_{yz}xz\left(x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](E_u^{(a)},1)$
** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{3\sqrt{5}iG_u z\left(x+iy\right)}{20}+\frac{\sqrt{15}iG_v z\left(x-iy\right)}{20}+\frac{\sqrt{15}G_{xy}z\left(x-iy\right)}{5}-\frac{\sqrt{15}iG_{xz}\left(x^2-4ixy-z^2\right)}{20}+\frac{\sqrt{15}iG_{yz}\left(4xy-iy^2+iz^2\right)}{20}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](E_u^{(a)},2)$
** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{3}iG_u z\left(x+iy\right)}{4}-\frac{3iG_v z\left(x-iy\right)}{4}+\frac{iG_{xz}\left(x^2-2y^2+z^2\right)}{4}+\frac{G_{yz}\left(2x^2-y^2-z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](E_u^{(a)},1)$
** symmetry

$$\frac{\sqrt{2}x^3}{2}-\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}+\frac{\sqrt{2}iy^3}{2}-\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{10}iG_u z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{8}+\frac{\sqrt{30}iG_v z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{24}$$
$$+\frac{\sqrt{30}G_{xy}z\left(4x^3+3ix^2y-3xy^2-3xz^2-4iy^3+3iyz^2\right)}{24}-\frac{\sqrt{30}iG_{xz}\left(x^4-4ix^3y-6x^2y^2+3ixy^3+3ixyz^2+6y^2z^2-z^4\right)}{24}$$
$$-\frac{\sqrt{30}iG_{yz}\left(3x^3y-6ix^2y^2+6ix^2z^2-4xy^3+3xyz^2+iy^4-iz^4\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](E_u^{(a)},2)$
** symmetry

$$\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}-\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{6}iG_u z\left(9x^3-12ix^2y-12xy^2-5xz^2+9iy^3-5iyz^2\right)}{24}-\frac{\sqrt{2}iG_v z\left(x^3-36ix^2y+36xy^2-13xz^2-iy^3+13iyz^2\right)}{24}$$
$$+\frac{7\sqrt{2}G_{xy}z\left(2x^3+3ix^2y-3xy^2-xz^2-2iy^3+iyz^2\right)}{24}+\frac{\sqrt{2}iG_{xz}\left(5x^4-14ix^3y-12x^2y^2-18x^2z^2+7ixy^3+21ixyz^2+4y^4-12y^2z^2+5z^4\right)}{24}$$
$$-\frac{\sqrt{2}G_{yz}\left(4x^4-7ix^3y-12x^2y^2-12x^2z^2+14ixy^3-21ixyz^2+5y^4-18y^2z^2+5z^4\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](E_u^{(b)},1)$
** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{3\sqrt{5}iG_u z\left(x-iy\right)}{20}-\frac{\sqrt{15}iG_v z\left(x+iy\right)}{20}+\frac{\sqrt{15}G_{xy}z\left(x+iy\right)}{5}+\frac{\sqrt{15}iG_{xz}\left(x^2+4ixy-z^2\right)}{20}-\frac{\sqrt{15}iG_{yz}\left(4xy+iy^2-iz^2\right)}{20}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](E_u^{(b)},2)$
** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$\frac{\sqrt{3}iG_u z\left(x-iy\right)}{4}+\frac{3iG_v z\left(x+iy\right)}{4}-\frac{iG_{xz}\left(x^2-2y^2+z^2\right)}{4}+\frac{G_{yz}\left(2x^2-y^2-z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](E_u^{(b)},1)$

** symmetry

$$\frac{\sqrt{2}x^3}{2}+\frac{3\sqrt{2}ix^2y}{4}-\frac{3\sqrt{2}xy^2}{4}-\frac{3\sqrt{2}xz^2}{4}-\frac{\sqrt{2}iy^3}{2}+\frac{3\sqrt{2}iyz^2}{4}$$

** expression

$$\frac{\sqrt{10}iG_uz\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{8}-\frac{\sqrt{30}iG_vz\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{24}$$
$$+\frac{\sqrt{30}G_{xy}z\left(4x^3-3ix^2y-3xy^2-3xz^2+4iy^3-3iyz^2\right)}{24}+\frac{\sqrt{30}iG_{xz}\left(x^4+4ix^3y-6x^2y^2-3ixy^3-3ixyz^2+6y^2z^2-z^4\right)}{24}$$
$$+\frac{\sqrt{30}iG_{yz}\left(3x^3y+6ix^2y^2-6ix^2z^2-4xy^3+3xyz^2-iy^4+iz^4\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](E_u^{(b)},2)$

** symmetry

$$-\frac{\sqrt{30}ix^2y}{4}+\frac{\sqrt{30}xy^2}{4}-\frac{\sqrt{30}xz^2}{4}+\frac{\sqrt{30}iyz^2}{4}$$

** expression

$$-\frac{\sqrt{6}iG_uz\left(9x^3+12ix^2y-12xy^2-5xz^2-9iy^3+5iyz^2\right)}{24}+\frac{\sqrt{2}iG_vz\left(x^3+36ix^2y+36xy^2-13xz^2+iy^3-13iyz^2\right)}{24}$$
$$+\frac{7\sqrt{2}G_{xy}z\left(2x^3-3ix^2y-3xy^2-xz^2+2iy^3-iyz^2\right)}{24}-\frac{\sqrt{2}iG_{xz}\left(5x^4+14ix^3y-12x^2y^2-18x^2z^2-7ixy^3-21ixyz^2+4y^4-12y^2z^2+5z^4\right)}{24}$$
$$-\frac{\sqrt{2}G_{yz}\left(4x^4+7ix^3y-12x^2y^2-12x^2z^2-14ixy^3+21ixyz^2+5y^4-18y^2z^2+5z^4\right)}{24}$$